

Taylor Cadwallader

Technical Game Designer

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HARD SKILLS

Design & Production: Gameplay design • Systems design • Technical design • UX design • Game feel design • Player controller design • Core mechanic design • Level/puzzle design • Progression design • Economy design • Quota design • Combat design • Enemy design • Multiplayer design • Network gameplay design • Procedural generation design • Tool design • Prototype design • Iterative design • Scope management • Feature cutting and simplification • Production planning • Lightweight project management • Scrum workflow support • Stand-up leadership • Task tracking • Cross-discipline coordination

Implementation & Programming: Unity development • Unreal Engine development • Custom C++ engine workflows • C# scripting • Unity Visual Scripting • Unreal Blueprints • Custom C# scripting layer work • Gameplay scripting • Feature implementation • Inspector-driven iteration • Gameplay debugging • Real-time particle system implementation • JSON-based data workflows • Data validation • Runtime system development • Tool scripting • WinForms development • Input handling systems • Serialization workflows • Runtime object spawning • Runtime mesh manipulation • Prefab-based workflows • ScriptableObject-driven architecture • Navmesh-based movement logic • Camera system implementation • Scene transition systems • UI-less gameplay feedback systems • Timer systems • Achievement tracking systems • Inventory systems • Interaction systems • Raycast-based interaction • Sphere-overlap suction mechanics • Physics-based controller development • Procedural dungeon generation • Multiplayer synchronization concepts • Server authority concepts

Tools & Workflow: Git • GitHub • SourceTree • Branching • Merging • Conflict resolution • Unity project setup • Source control onboarding • Jira • ClickUp • Trello • Google Sheets as a data store • Playtesting documentation • Engine workflow documentation • Design request tracking • Custom editor tools • QA-style bug reporting • Usability issue documentation • Spreadsheet-driven production communication

Art, UX, & Presentation Support: Blender • Maya • Figma • Miro • Photoshop • DaVinci Resolve • Portfolio presentation support • Prototype presentation setup • Visual readability tuning • Environmental readability support • Particle and lighting coordination • Camera readability tuning • Audio-feedback iteration support

Testing & Iteration: Structured playtesting • Playtest report writing • Feedback analysis • Player behavior observation • Clarity iteration • Usability iteration • Mechanic communication tuning • Rapid prototyping • Iteration under time constraints • Cross-build review • Bug discovery and triage • System refinement based on feedback

SOFT SKILLS

Collaboration & Communication: Clear communication • Collaborative problem solving • Cross-disciplinary communication • Idea sharing • Feedback seeking • Course correction • Team alignment • Production clarity • Explaining technical/design intent to others • Working with programmers, artists, and audio designers • Working on small teams • Working on large teams • Acting as a bridge between design and implementation

Leadership & Ownership: Leadership • Design leadership • Acting as lead programmer in practice • Taking ownership of implementation • Running SCRUM meetings • Keeping teams on track • Responsibility under ambiguity • Initiative • Ownership of core systems • Supporting teammates who are new to tools or engines

Problem Solving & Thinking Style: Logical thinking • Systems-oriented thinking • Structured problem solving • Strong polish focus • Iterative improvement mindset • Technical curiosity • Adaptability • Tradeoff evaluation • Scope awareness • Constraint-based design • Practical decision-making • Debugging mindset • Readability-first thinking • Player-centric thinking

Work Style & Process: Comfortable with fast-moving environments • Comfortable with constrained environments • Comfortable with rapid prototyping • Comfortable revising work based on playtest feedback • Willingness to simplify or cut features • Persistence through messy technical situations • Self-directed learning • Comfort with unfinished or evolving tools • Ability to transition between prototype and final implementation • Ability to work across disciplines without being hand-held

PROJECTS

Game Designer (Support Role)

Feb 2026

AntiMasking • 3D Endless Runner • Unity • Game Jam

Team of 10

- Contributed during the early brainstorming phase of a 48-hour Global Game Jam project with a ten-person team.
- Helped keep the concept coherent and technically achievable within the jam constraints.
- Connected ideas together, flagged scope and design issues, and helped narrow the team toward a realistic direction.
- Helped troubleshoot level whiteboxing issues and adjusted turning triggers for the auto-running character's corner movement.
- Made light documentation updates by adding references and clarifying information to keep the team aligned.

Gameplay Designer & Programmer

Dec 2025 - Feb 2026

Trash • 3D Top-Down Extraction Game • Unity • Academic

Team of 3

- Serving as my final academic project at DigiPen, developed by a team of three and still in progress.
- Built a 3D top-down extraction-style game where the player vacuums up trash and materials and fires them into collectors to meet quotas.
- Took on the primary gameplay role and designed and implemented nearly all core systems.
- Built the navmesh-based player controller, vacuum gun, suckable object framework, collectors, levers, trash piles, inventory system, quota system, controller support, camera orbit controls, and ship transitions.
- Added game feel features including screenshake, FOV shifts, and rumble feedback.
- Implemented the navmesh-driven controller with mouse-facing rotation, directional dash, and "fake" movement via fed position arrays for transitions and possible cinematics.
- Built the vacuum gun as a layered mechanic with sphere-overlap suction, spiral force, adjustable strength, inventory insertion, clogging behavior, and automatic firing with cooldown ramping.
- Designed precision-based interaction for levers and collectors using forward raycast targeting and sustained input feedback.
- Built a ScriptableObject-driven framework for suckable objects with adjustable physics, spiral interaction, break-apart behavior, large-state behavior, firing force, cooldown ramping, and spawn-on-break logic.
- Designed the quota system around a percentage of total spawned trash, an in-world time limit, early submission rewards, strikes, and soft/hard failure states.
- Currently developing a charge mechanic that condenses full inventory slots into a single large projectile for faster deposits and deeper decision-making.

Tools Programmer & Software Developer

Nov 2025 - Dec 2025

Input Replay Analyzer • Input Analysis & Replay Tool • C# WinForms • Academic

Solo project

- Built a standalone WinForms tool over four weeks to analyze recorded player input logs from playtest sessions.
- Designed it for QA testers, user researchers, and designers who needed a clearer way to inspect player behavior and reproduce bugs.
- Added timeline-based log visualization with filtering and trimming by input type or time range.
- Supported exporting cleaned logs so specific moments could be isolated and shared.
- Implemented full input replay functionality using original timestamps to reproduce sessions locally.
- Built a profile system where users could define custom input maps and save them as JSON for different games and control schemes.
- Added an executable focus check so recording only happened while the target app was active, preventing unrelated input capture and helping with privacy/data accuracy.
- Strengthened my skills in tool development, input handling, data serialization, and standalone debugging utilities outside a game engine.

Tools Programmer & Technical Designer

Sep 2025 - Oct 2025

3D Sprite Layerer • Unity Editor Tool • C# • Academic

Solo project

- Created a custom Unity editor tool over about three weeks to automate layered sprite prefab creation.
- Built it to speed up environment creation for a 2.5D prototype that used stacked sprites in 3D space for a paper-like depth effect.
- Replaced a manual workflow involving sprite duplication, placement, and value tweaking with automated prefab generation.
- Supported arrays of sprites with configurable spacing, angles, ordering, random positional offsets, variant generation, and save-location control.

- Designed the tool to be useful for future collaborators, especially artists, so they could generate usable props without manual scene assembly.
- Improved early prototype workflow by reducing environment setup time and keeping focus on gameplay systems.

Systems Designer & Gameplay Programmer

Aug 2025

Line 9 • 3D Puzzle Inspection Game • Unity • Game Jam

Team of 6

- Completed a 72-hour GMTK Game Jam project on a remote six-person team.
- Helped consolidate team brainstorming into a single direction everyone felt good about.
- Took on more hands-on implementation because the team had fewer programmers than BLOOPS.
- Designed and implemented the core anomaly system, including randomized doll features and traits flagged as correct or faulty.
- Implemented a difficulty scaling system where anomalies became subtler as the player improved.
- Built inspection stations using layer masks to simulate different viewing lenses while reusing a shared system.
- Created the UV station, magnifier station, and X-ray station, each exposing different anomaly types.
- Focused on building a flexible framework that could be iterated on and re-skinned without overwhelming the 72-hour scope.

Systems Designer & Gameplay Designer

Summer 2025

Strip Miner • Voxel Mining Prototype • Unity

Team of 3

- Worked on a small three-person summer project built around the fantasy of strip mining in a voxel-based world.
- Focused on gameplay prototyping and system design while the programmer built the voxel foundation.
- Implemented and tested the early mining loop, deposit flow, and upgrade structure.
- Wrote the full game design document covering upgrade systems, quota progression, helper items, and a roguelike layer structure.
- Designed the core loop around mining ore, depositing it for credits, upgrading tools, and pushing deeper into underground layers.
- Focused on pacing and motivation, including slow early progression and later power fantasy escalation.
- Designed systems such as quota requirements, helper tools, and depth-based unlocks to reinforce forward momentum.

Solo Developer & Game Feel Designer

Mar 2025 - Apr 2025

Depth Strider • First-Person Kinesthetic Experience • Unity • Academic

Solo project

- Completed a five-week DES365 prototype focused on deep-sea diver movement feel and immersion.
- Built a custom Unity player controller with Cinemachine for smooth look rotation, camera bob, and subtle screen shake.
- Tuned movement to feel slow, heavy, and resistant, with gradual acceleration to simulate underwater movement.
- Implemented delayed head rotation so the camera lagged slightly behind player input, reinforcing the feel of a bulky diving suit.
- Added angle limits to the camera and helmet to keep the player view grounded and restricted.
- Built an underwater ravine environment that naturally guided the player forward without explicit navigation systems.
- Used a distant ship with blinking light as a persistent visual goal that grew larger as the player approached.
- Added post-processing, fog, a custom skybox, floating particles, and breathing bubble effects to reinforce the underwater atmosphere.
- Prioritized audio and ambience early, including underwater ambience and breathing sounds.

QA Tester & Technical Designer

Jan 2025 - Apr 2025

NiteLite • 3D Custom C++ Engine with C# Scripting Layer • Academic

Team of 9

- Worked on a custom 3D C++ engine with a custom C# scripting layer alongside another designer.
- Spent substantial time playtesting engine features, reporting bugs, documenting usability issues, and suggesting missing workflow improvements from a designer's perspective.
- Built the player navigation and scene transition system for the final engine demo.
- Implemented a basic first-person controller and interactive doors that connected prototype showcase scenes.
- Created a non-Euclidean transition effect by controlling scene loads and teleporting the player while the camera was obscured.
- Built a standardized showcase room template with a viewing window and interaction point to present prototypes in a consistent format.
- Contributed to design-focused usability feedback that improved the engine workflow for the rest of the team.

Gameplay Designer & Programmer

Jan 2025

- Completed a 96-hour Global Game Jam project on a nine-person multidisciplinary team.
- Helped guide the team toward a shared concept during brainstorming after the theme reveal.
- Assisted one programmer who was new to Unity by helping them navigate the engine and troubleshoot issues.
- Handled most of the 3D gameplay asset implementation, connecting art assets to systems and ensuring prefabs worked correctly in-game.
- Designed the mechanics that changed bubble trajectory and structured the puzzle flow.
- Implemented the camera perspective swap toggle, randomized floor tile generator, counter prop spawner, and an early inventory prototype.
- Focused on making sure the core mechanics worked and the team stayed aligned within a tight deadline.

Solo Developer & Systems Designer**Jan 2025 - Feb 2025**

Forgeborne • First-Person Action Prototype • Unity • Academic

Solo project

- Completed a five-week solo prototype for DES365 Game Feel in Unity.
- Built a first-person action experience centered on electromagnetism and adaptive weapon forging.
- Created a runtime mesh-splitting system that broke assigned models into individual triangles to simulate metal shrapnel.
- Built a reform animation pipeline that teleported, spun, and reassembled shrapnel into weapons, then dispersed it again during attacks.
- Designed three weapon archetypes, each with three attack types, all tied into the same modular shrapnel system.
- Pivoted away from a handcrafted primitive-shape approach when it became clear it would not scale within the five-week timeline.
- Focused iteration on stabilizing and refining the electromagnetism system.
- I learned that my strongest work tends to come from building flexible systems first and expressing them through multiple mechanics.

UX & Combat Designer**Sep 2024 - Dec 2024**

Knight Light • 3D Action Adventure • Unreal Engine • Academic

Team of 25

- Joined an existing senior team and helped continue development on a 3D third-person action-adventure game in Unreal.
- Brought in primarily for combat design and UX work.
- Played the latest build and documented what worked, what needed improvement, and what felt inconsistent or unclear.
- Created a large feedback document to identify key problem areas from a UX and gameplay perspective.
- Built and maintained a large Design Requests spreadsheet that broke down actionable tasks across programming, art, audio, and VFX.
- Added context, reasoning, and implementation notes to requests so team members understood both the problem and the intended outcome.
- Wrote design documents proposing improvements to player combat, enemy behavior, and new enemy concepts.
- Tuned player and enemy variables and helped rebalance the wave system in Unreal.
- Contributed to a large team of around 25 people by strengthening cohesion and production clarity.

Recognition: Featured on DigiPen's Game Gallery and Steam.**Systems Designer & Gameplay Programmer****Summer 2024**

AGP (Alien Game Project) • Multiplayer Extraction Prototype • Unity

Team of 2

- Helped build a Lethal Company-style co-op prototype over the summer with one other designer.
- Gained first serious experience with multiplayer gameplay architecture, including synchronization, shared game states, and server authority decisions.
- Implemented a physics-based player controller and a networked interaction/inventory system for picking up and dropping items.
- Learned Scriptable Object workflows and used them to define item data.
- Built a node-based procedural dungeon generator that connected room prefabs through doorways and randomly selected and rotated rooms during expansion.
- Designed rules for drop-off zones, including guaranteed spacing and a seed-based active insertion point.
- Implemented drop-pod deployment and ship extraction sequences synchronized for all players.
- Helped explore a larger refuge-and-escape concept centered on deploying, looting, and extracting before the ship fled again.

UX & Systems Designer

Sep 2023 - May 2024

KnightLight • 2D Adventure Platformer • Custom C++ Engine • Academic

Team of 7

- Served as Design Lead on a multidisciplinary two-semester project built on a custom C++ engine.
- Created frequent Unity prototypes, sometimes weekly, to test gameplay concepts, player abilities, lighting interactions, and movement feel while the engine was still being built.
- Helped scale the project down from a larger combat/metroidvania concept to a puzzle-focused game centered on light.
- Redesigned the player into two interconnected parts, the Knight and the Light, creating a mechanic where the Light could move independently while the Knight remained behind.
- Designed and implemented the Flash ability, lamp/door pairing progression system, and checkpoint torches that functioned as save points and recall points.
- Iterated on visual and audio feedback after playtests showed players were confused about the armor recall system.
- Created the Signiflyers collectible system, which doubled as subtle visual guidance in dark spaces.
- Implemented most Steam achievements by building tracking hooks across multiple gameplay systems.
- Built the pipe traversal system using directional tags, giving the level designer flexibility in puzzle construction.
- Helped reinforce readability and atmosphere through lighting, particle work, and a proximity-based red glow on spike hazards.
- Conducted regular playtests across both semesters and authored numerous detailed reports on player behavior, confusion points, and usability issues.
- Co-authored an Engine Design Guide documenting workflows for future designers working in the custom engine.
- Transitioned from Unity prototyping to direct implementation in the custom engine once it became stable.

Recognition: Highlighted on DigiPen's Game Gallery and Steam (110+ positive reviews).

Systems Designer & Programmer

Summer 2023

SGP (Streamer Game Project) • Twitch-Integrated Interactive Prototype • Unity

Team of 2

- Built a prototype exploring how Twitch chat could directly drive an in-engine Unity experience.
- Designed and implemented a system that spawned avatars in-scene based on live chat activity, with floating nametags tied to usernames.
- Added lifetime timers to avatars, resetting them and triggering a hop animation whenever the same chatter sent another message.
- Prototyped networking so chatters could connect directly to the game and control their own avatars in first person.
- Integrated a public Google Sheet as a persistent data store keyed by username to support inventories, currency, and cosmetic unlocks.
- Implemented a command-driven customization system where users could change avatar color using a "!color" command with validated and saved values.
- Built a dynamic emote parsing system that detected chat emotes and spawned particle effects using the emote image as the sprite.
- Worked on modular live-data pipelines and readable feedback loops connecting community interaction to gameplay space.

Systems Designer & Programmer

Jan 2023 - Apr 2023

Project Nasu • 2D Action Platformer • Unity • Academic

Team of 3

- Served on a three-designer team and naturally stepped into a leadership role after coming from a technical school background.
- Set up the Unity project, introduced the team to source control, helped establish workflow, and ran SCRUM meetings to keep production on track.
- Took on most of the implementation work and effectively acted as the lead programmer while still contributing to design.
- Designed, prototyped, and implemented the player controller, collectibles, hazards, enemies, and a speedrun timer with time tracking.
- Built readable hazard tools for level design, including three laser types: a red laser disabled by shooting a level button, a green laser that blinked on an interval, and a blue laser that rotated within a 90-degree range.
- Implemented enemy behaviors including a rat that patrolled between two points, a rat spawner that respawned rats after death, and a mutated hamster that combined patrol behavior with laser attacks.
- Gained first-hand experience fully owning both design and implementation on a team.

Recognition: Showcased in Foundry10's K-12 audio workshops; awarded team honorarium.

EDUCATION

Bachelor of Arts in Game Design

DigiPen Institute of Technology, WA

Apr 2026

Dean's Honor List: Spring 24 & Fall 23

Video Game Designer

Sno-Isle TECH Skills Center, WA

Sep 2022

For 1,200 hours